



Big Data Visual Analytics (CS 661)

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Study Materials for Lecture 13

- <https://jdstorey.org/fas/index.html>
- <https://online.stat.psu.edu/stat500/lesson/0>
- A Gentle Tutorial of the EM Algorithm and its Application to Parameter Estimation for Gaussian Mixture and Hidden Markov Models
- EM Algorithm:
https://stephens999.github.io/fiveMinuteStats/intro_to_em.html

Random Variables and Distributions

Random Variable

- Let S be a sample space of an experiment
 - S is associated with a probability measure P
- A random variable X is a real valued function on S
- Key property: It is a function whose values have probabilities attached with it

Random Variable: Example

- Let us flip a fair coin three times
- Sample space $S = \{hhh, hht, hth, htt, thh, tht, tth, ttt\}$
- Assume X is a function on S , so that X is the number of heads (h)
- So, we have,
 - $\{hhh \rightarrow 3, hht \rightarrow 2, hth \rightarrow 2, htt \rightarrow 2, thh \rightarrow 2, tht \rightarrow 2, tth \rightarrow 1, ttt \rightarrow 0\}$
- X is a random variable

Random Variable: Example

- We can answer questions like:
 - $P(X=0) = P(\text{ttt}) = 1/8$
 - $P(X = 1) = P(\text{htt}) + P(\text{tht}) + P(\text{tth}) = 3/8$
 - $P(X = 2) = P(\text{hht}) + P(\text{hth}) + P(\text{thh}) = 3/8$
 - $P(X = 3) = P(\text{hhh}) = 1/8$
- We can tabulate it:

X	0	1	2	3
$P(X = x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

Random Variable (RV): Example

- Rolling a fair die
- Assume a RV: X = the number that comes up
- X takes values 1,2,3,4,5,6 with probability $1/6$

X	1	2	3	4	5	6
$P(X = x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

Discrete and Continuous Random Variable

- A random variable is said to be discrete if its set of possible values is a discrete set
 - Example: Rolling a fair die and measuring the value that shows up
- A random variable is said to be continuous when it can assume an uncountable number of values
 - Example: Depth of a pool, height of all the males, etc.

Expected Value and Variance of a Discrete RV

- Expected Value (mean):

$$E(X) = \sum x_i * p(x_i), \quad p(x) = PMF$$

- Variance:

$$Var(X) = \sum (x - E(X))^2 * p(x)$$

- Standard Deviation:

$$SD(X) = \sqrt{Var(X)}$$

Expected Value and Variance of a Continuous RV

- Expected Value (mean):

$$E(X) = \int_{-\infty}^{\infty} x * f(x) dx, \quad f(x) = PDF$$

- Variance:

$$Var(X) = \int_{-\infty}^{\infty} (x - E(X))^2 * f(x) dx$$

- Standard Deviation:

$$SD(X) = \sqrt{Var(X)}$$

Probability Distribution Function

- **A probability distribution function** is a mathematical function that provides probabilities of occurrence for the possible outcomes of a random variable
- **Probability Mass Function (PMF)**: The probability distribution of a discrete random variable is called probability mass function
- **Probability Density Function (PDF)**: The probability distribution of a continuous random variable is called probability density function

Probability Distribution Function: Properties

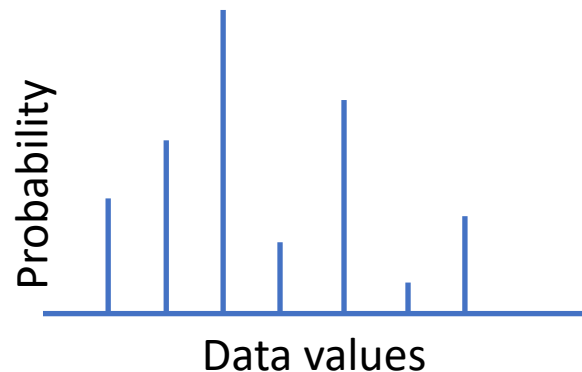
- Discrete case: PMF

- $p(x) = P(X = x)$

1. $p(x) \geq 0$

2. $\sum_{\text{all possible } x} p(x) = 1$

3. $p(x) = 0$ for all x outside a discrete range

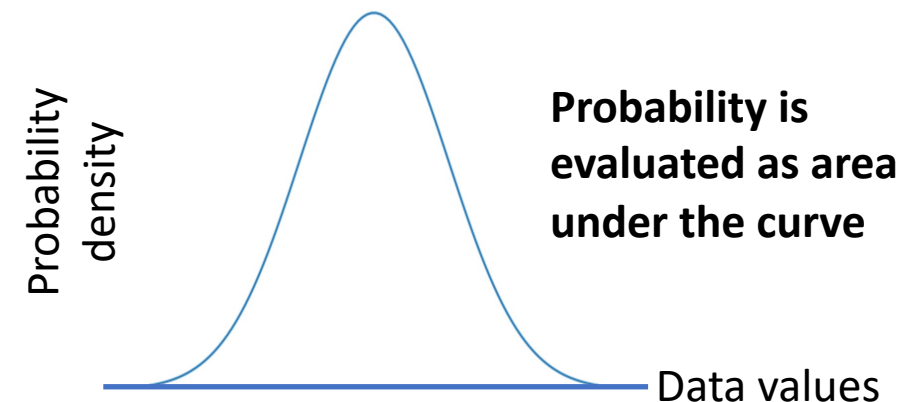


- Continuous case: PDF

- $f(x)$

1. $f(x) \geq 0$

2. $\int_{-\infty}^{\infty} f(x) dx = 1$



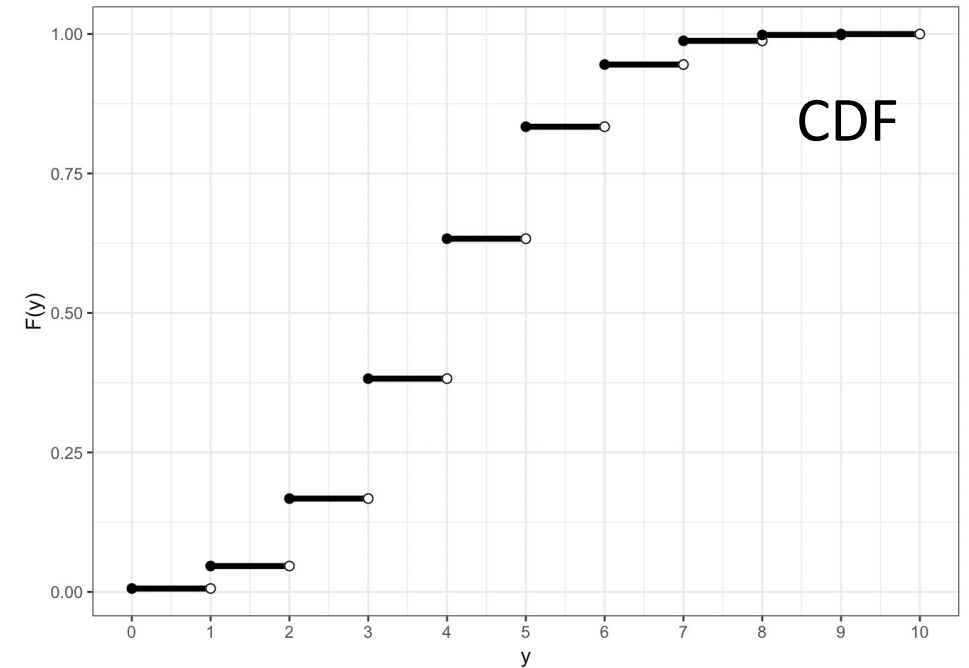
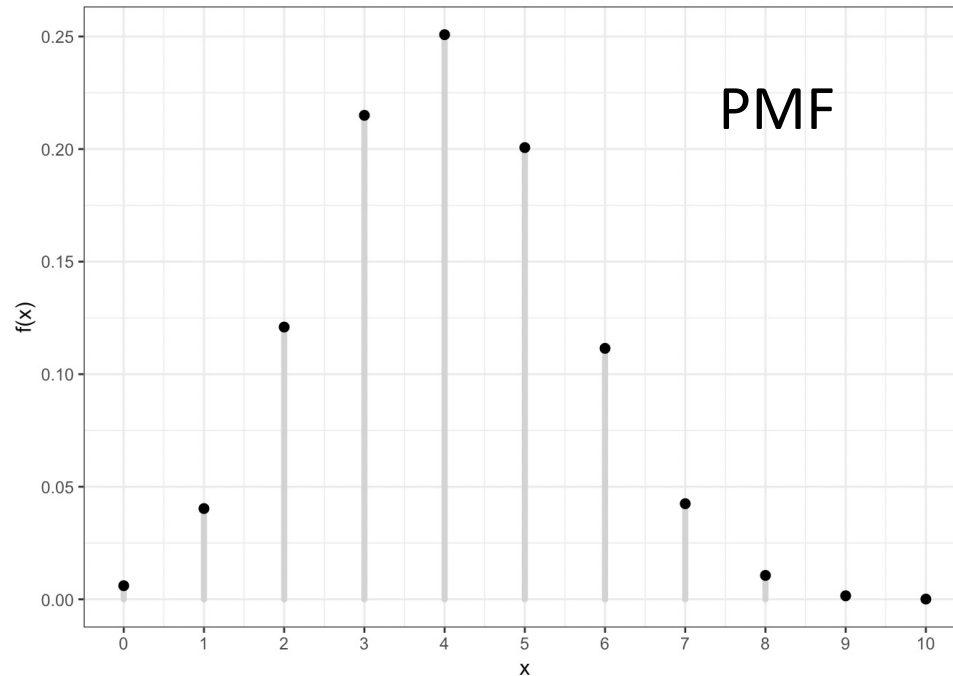
$P(x=c) = 0$ The probability that x takes on any individual value is zero. The area below the curve between $x=c$ and $x=c$ has no width, and therefore no area.

Cumulative Distribution Function (CDF)

- Discrete RV: Non decreasing function

$$F_X(x) = p(X \leq x) = \sum_{x_i \leq x} p(x_i)$$

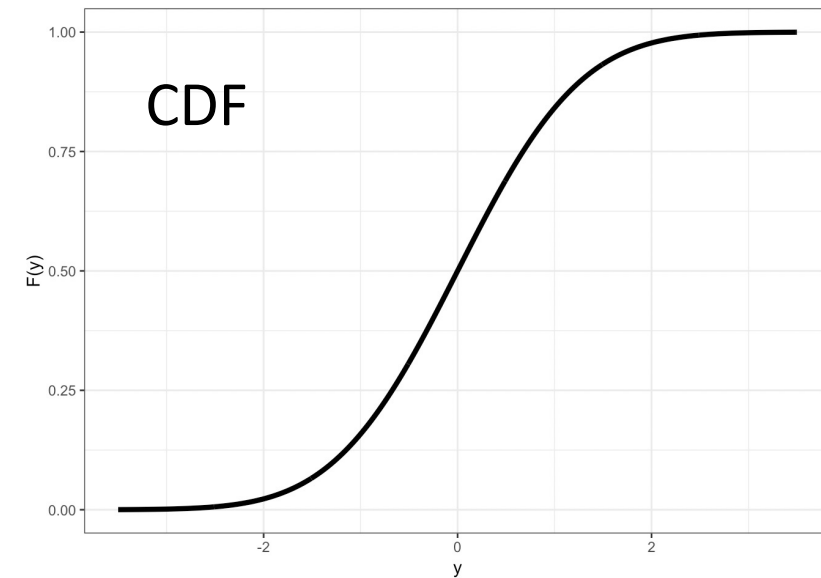
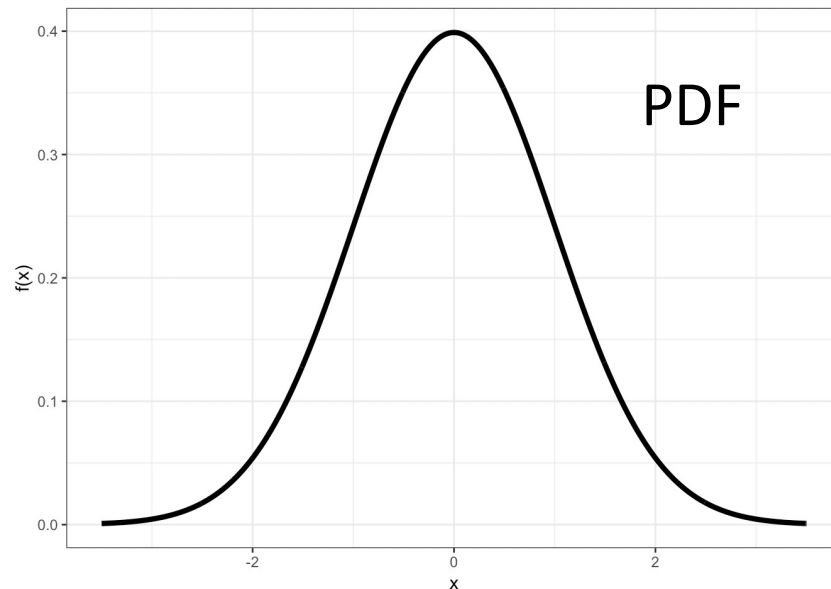
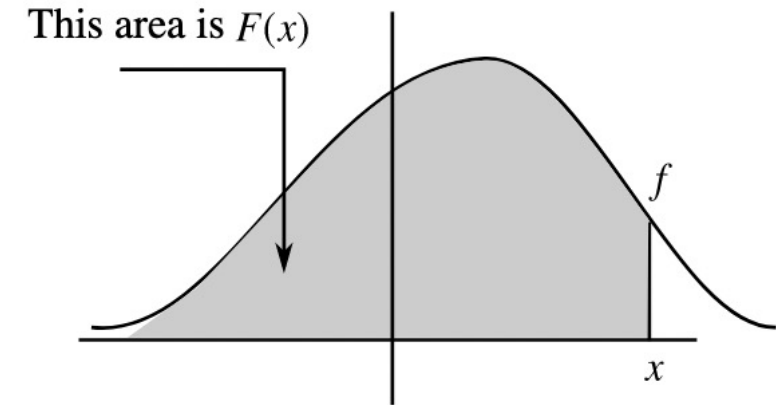
**CDF is a right continuous function
for discrete RV**



Cumulative Distribution Function (CDF)

- Continuous RV: Non decreasing function

$$F_X(x) = \int_{-\infty}^x f(x)dx$$



**CDF is a
continuous
function
here**

Discrete: Uniform Distribution

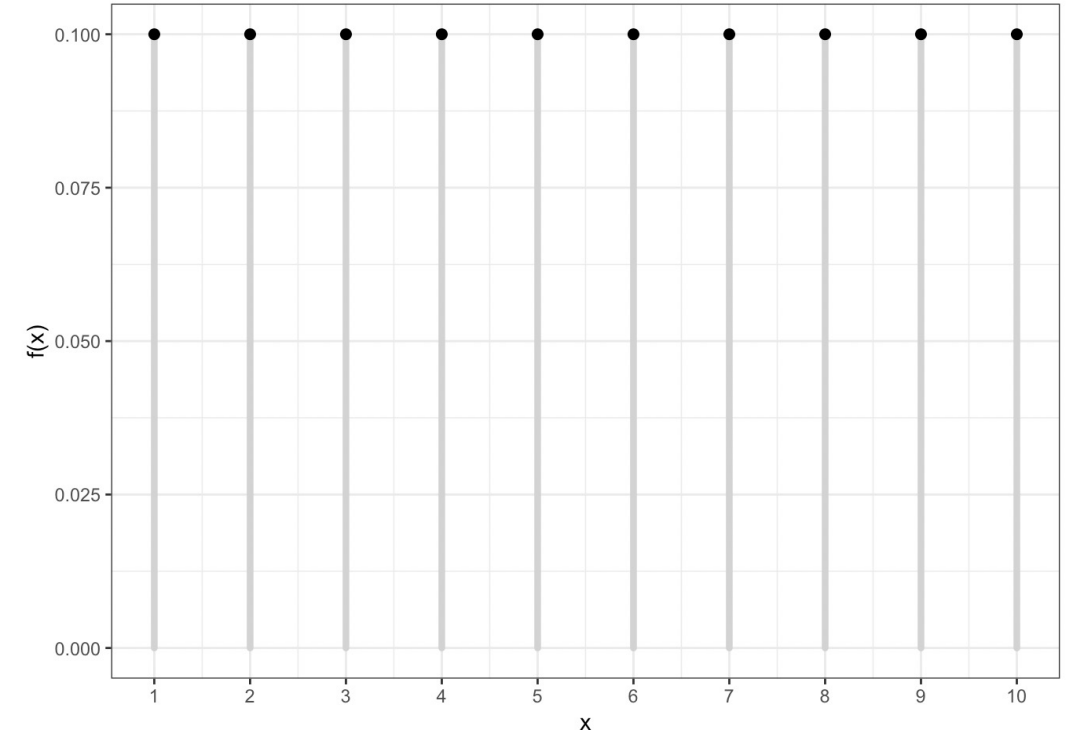
- Distribution assigns equal probabilities to a finite set of values

$$X \sim \text{Uniform}\{1, 2, \dots, n\}$$

$$\mathcal{R} = \{1, 2, \dots, n\}$$

$$f(x; n) = 1/n \text{ for } x \in \mathcal{R}$$

$$\mathbb{E}[X] = \frac{n+1}{2}, \text{Var}(X) = \frac{n^2-1}{12}$$



Continuous: Exponential Distribution

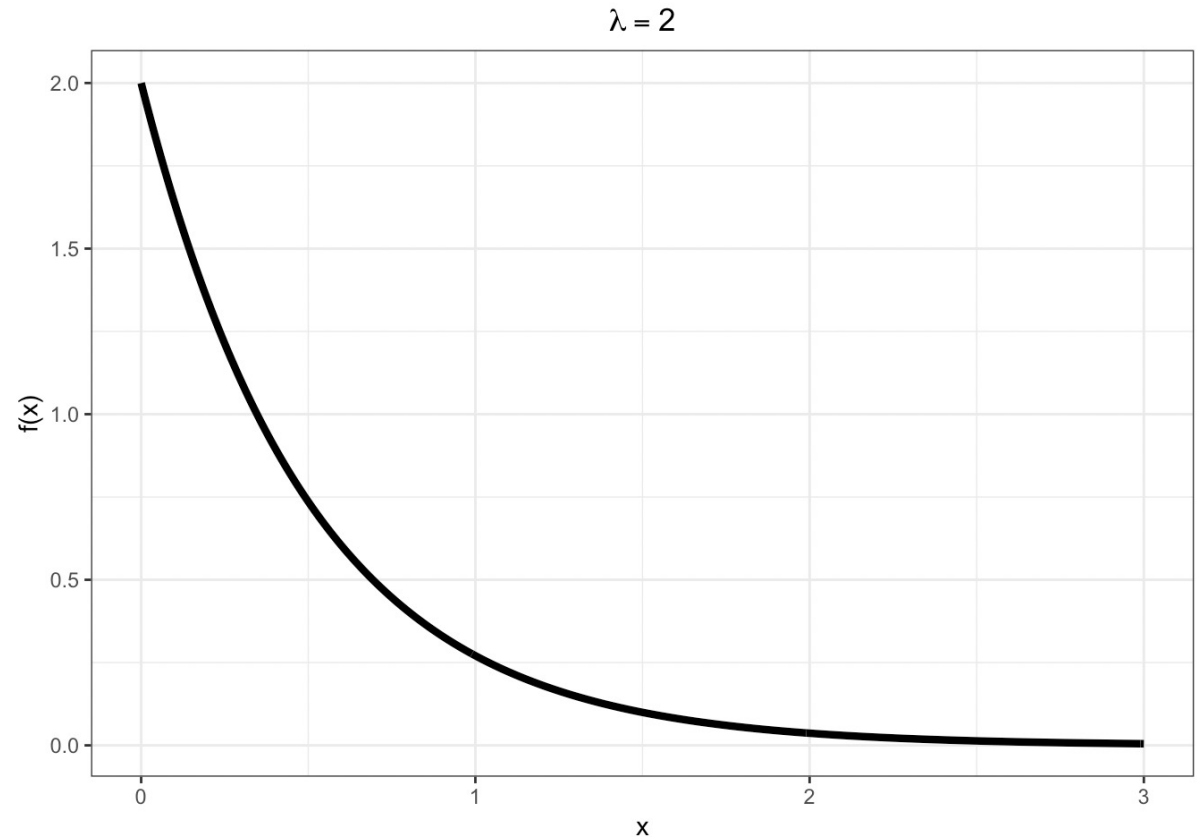
$$X \sim \text{Exponential}(\lambda)$$

$$\mathcal{R} = [0, \infty)$$

$$f(x; \lambda) = \lambda e^{-\lambda x} \text{ for } x \in \mathcal{R}$$

$$F(y; \lambda) = 1 - e^{-\lambda y} \text{ for } y \in \mathcal{R}$$

$$\mathbb{E}[X] = \frac{1}{\lambda}, \text{ Var}(X) = \frac{1}{\lambda^2}$$



Continuous: Beta Distribution

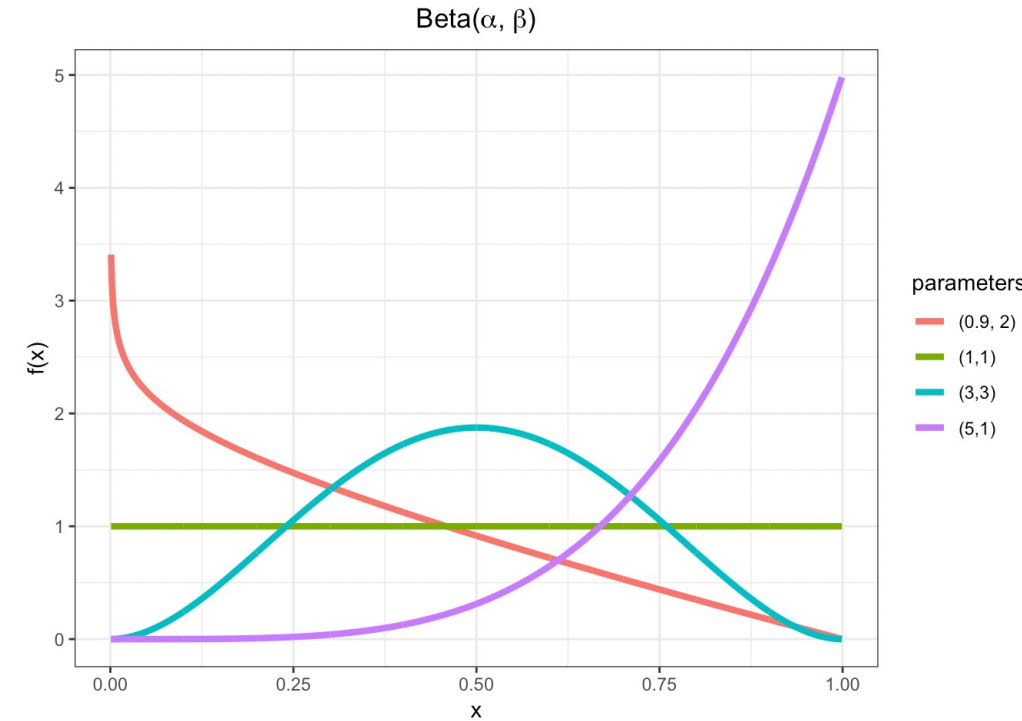
$X \sim \text{Beta}(\alpha, \beta)$ where $\alpha, \beta > 0$

$$\mathcal{R} = (0, 1)$$

$$f(x; \alpha, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1} \text{ for } x \in \mathcal{R}$$

where $\Gamma(z) = \int_0^\infty x^{z-1} e^{-x} dx$.

$$\mathbb{E}[X] = \frac{\alpha}{\alpha + \beta}, \quad \text{Var}(X) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$$



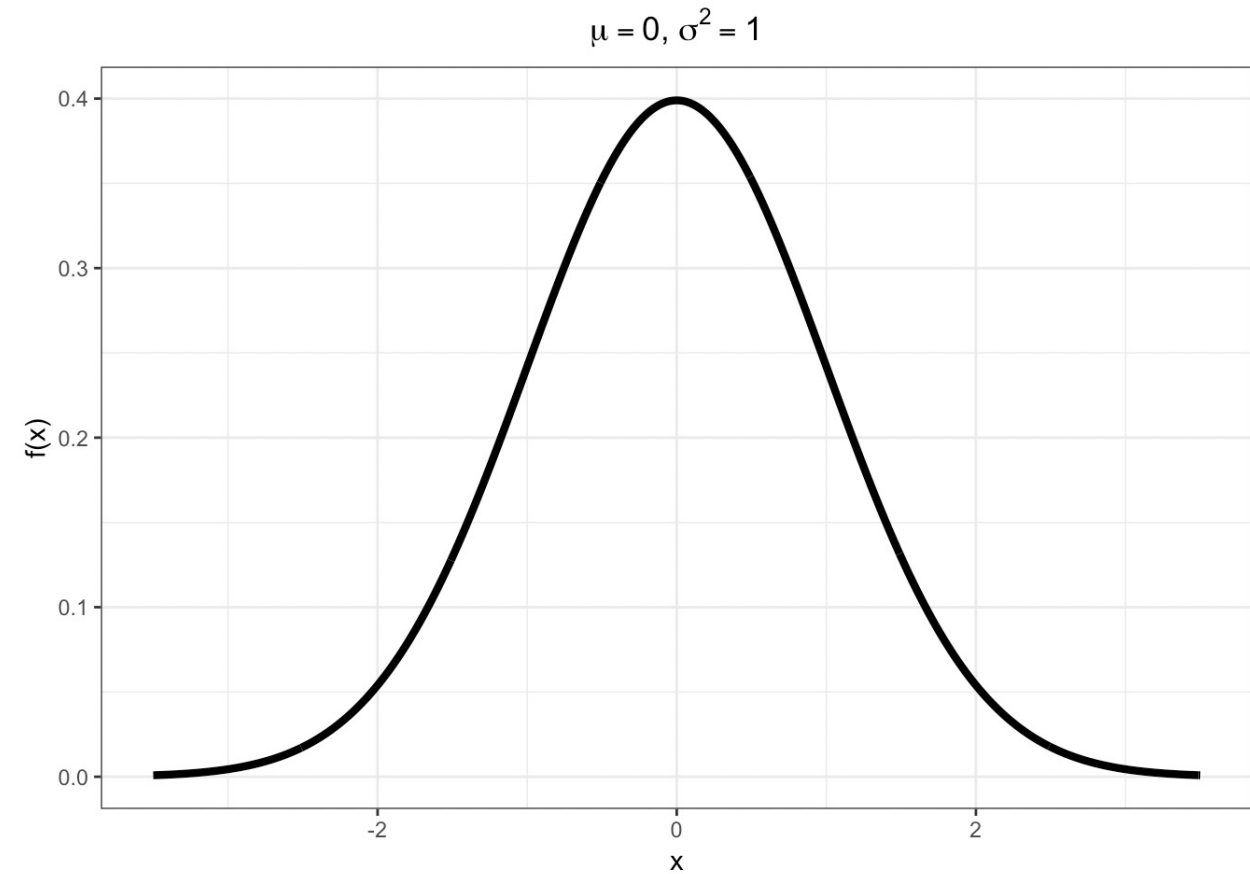
Continuous: Normal (Gaussian) Distribution

$$X \sim \text{Normal}(\mu, \sigma^2)$$

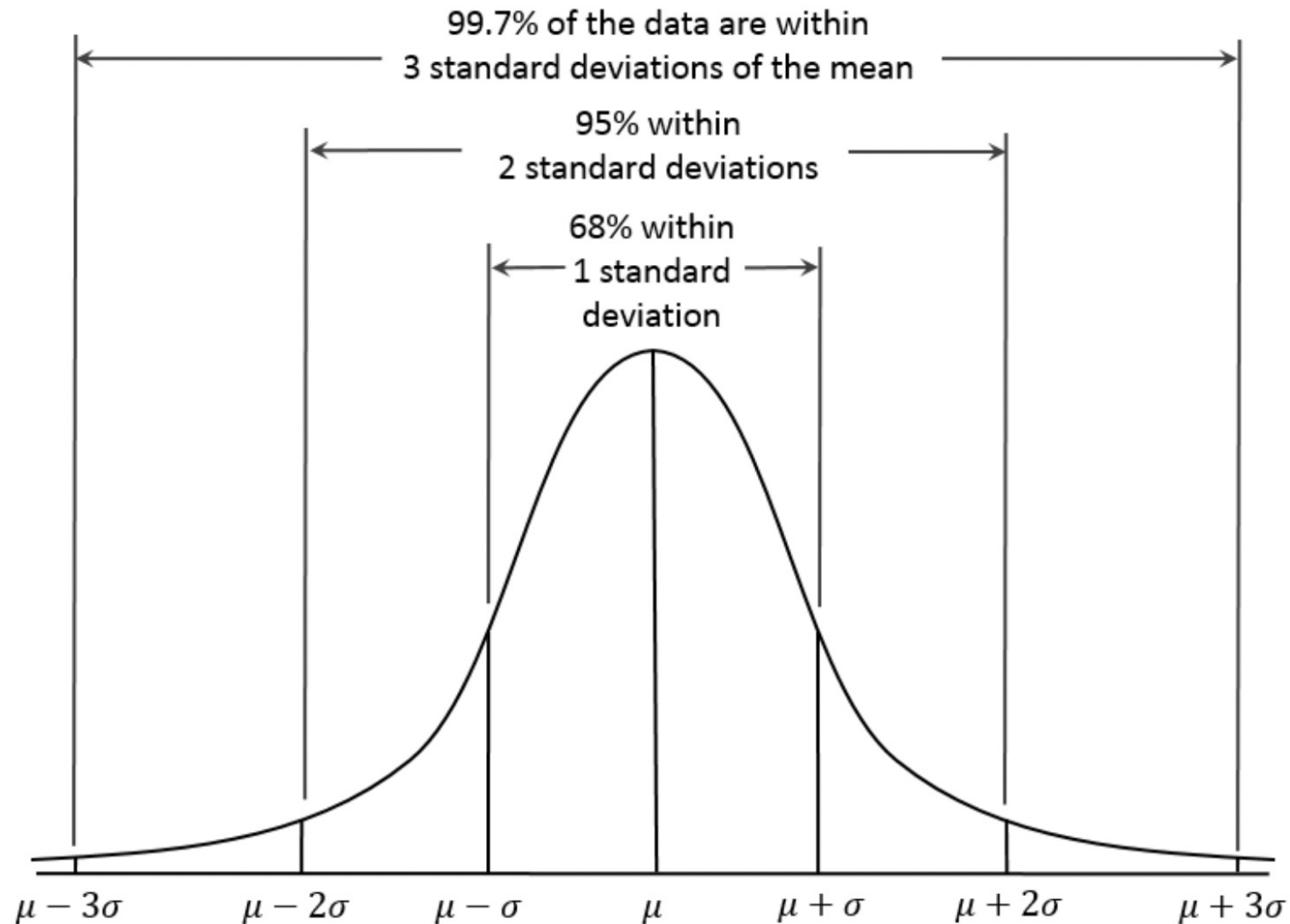
$$\mathcal{R} = (-\infty, \infty)$$

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \text{ for } x \in \mathcal{R}$$

$$\mathbb{E}[X] = \mu, \text{ Var}(X) = \sigma^2$$



Reading a Normal (Gaussian) Distribution



Continuous: Standard Normal Distribution

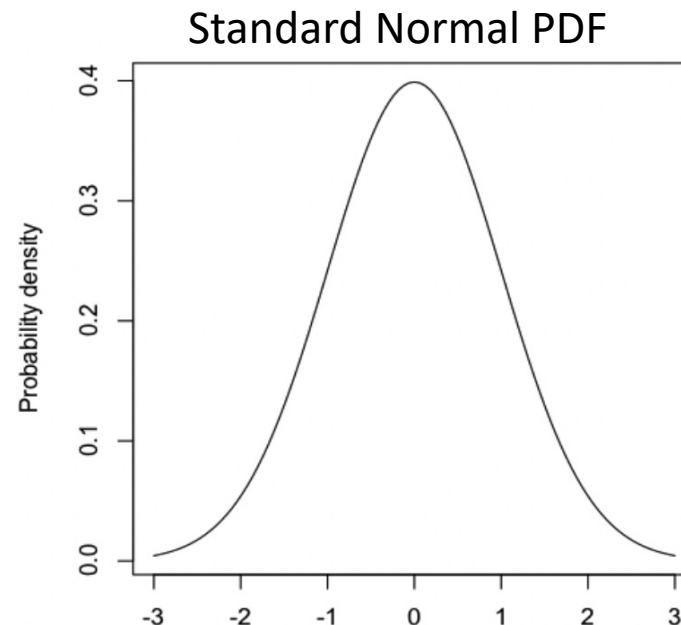
- It is the normal distribution with a mean equal to 0 and a standard deviation (also variance) equal to 1
- The standard normal distribution is often abbreviated to Z. It is frequently used to simplify working with normal distributions.

$$Z = \frac{X - \mu}{\sigma}$$

Continuous: Standard Normal Distribution

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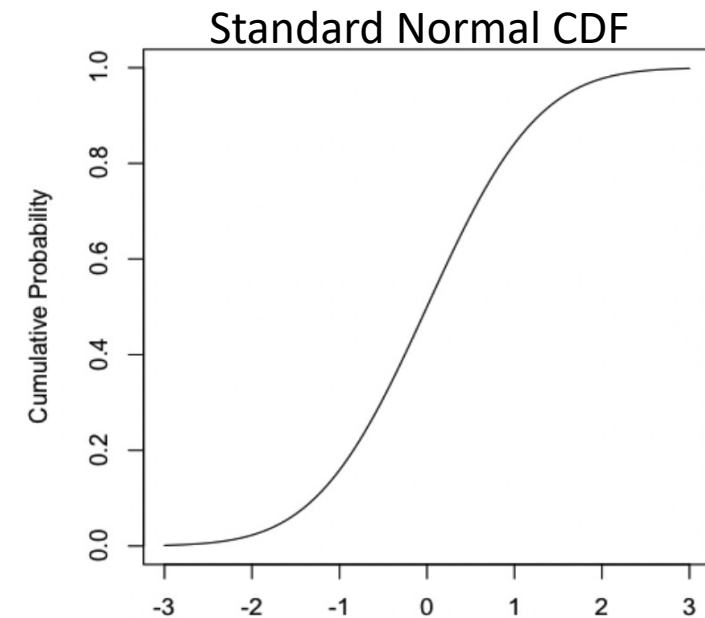
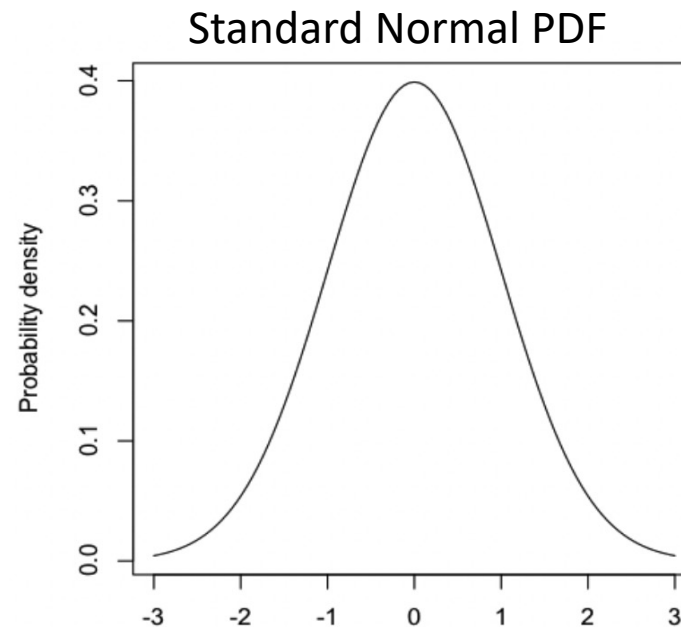
$$Z = \frac{X - \mu}{\sigma}$$



Continuous: Standard Normal Distribution

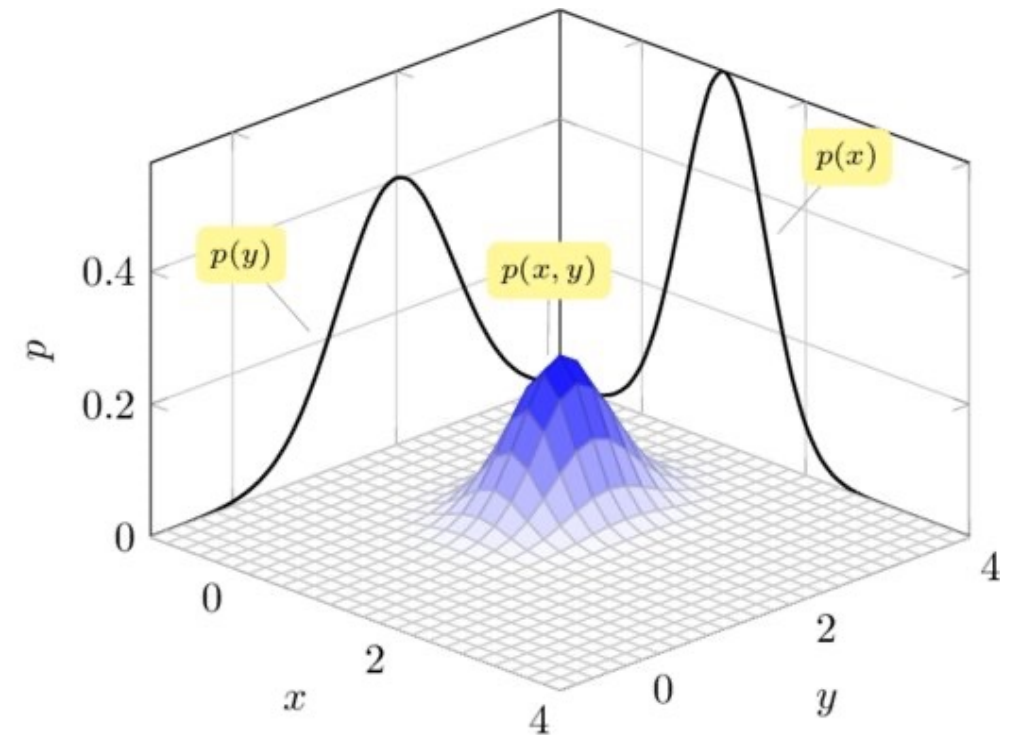
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$$Z = \frac{X - \mu}{\sigma}$$



Joint Probability Distribution Function

- If we have multiple random variables, defined over the same probability space S , then the joint probability distribution is the distribution function that is defined over all possible event combinations of all the random variables
- Joint probability density function for two continuous random variables X and Y can be represented as $f_{XY}(x, y)$



Joint Probability Distribution Function

- The concept of joint probability distribution function is generalizable and goes beyond two variables: $f_{X_1 X_2 X_3 \dots X_n}(x_1, x_2, x_3, \dots x_n)$
- For two variable case, $f_{XY}(x, y)$ must be a non-negative function and the following must hold:

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x, y) dx dy = 1$$

- Joint Cumulative Distribution function (CDF)

$$F_{XY}(x, y) = P(X \leq a, Y \leq b) = \int_{-\infty}^a \int_{-\infty}^b f_{XY}(x, y) dx dy$$

Marginal Probability Distribution Functions

- From the joint probability distribution function, we can find the marginal probability distributions by integrating the joint distribution function $f_{XY}(x, y)$

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy, \text{ for all } x$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y) dx, \text{ for all } y$$

- Marginal distribution functions (also known as univariate distributions) are probability distribution functions of individual random variables

Independence

- The continuous random variables are statistically independent if their joint probability distribution function factors into a product of their marginal distributions

$$f(x_1, x_2, x_3, \dots, x_n) = f(x_1)f(x_2)f(x_3) \dots f(x_n)$$

Conditional Probability and Bayes' Rule

- Conditional probability: It is the probability of an event given another event has occurred

$$f_{X|Y=y}(x) = \frac{\Pr(\{X = x\} \cap \{Y = y\})}{\Pr(Y = y)} = \frac{f_{X,Y}(x, y)}{f_Y(y)}$$

- Bayes' Rule:

$$f_{X|Y=y}(x) = \frac{f_{Y|X=x}(y) * f_X(x)}{f_Y(y)}$$

$f_{X|Y=y}(x)$ = Conditional probability of $X = x$ given $Y = y$. This is also called posterior probability

$f_{Y|X=x}(y)$ = Conditional probability of $Y = y$ given $X = x$. This is called likelihood

$f_X(x)$ = marginal of X , also the prior probability of $X = x$

$f_Y(y)$ = marginal probability of Y

Representations of Distribution Functions

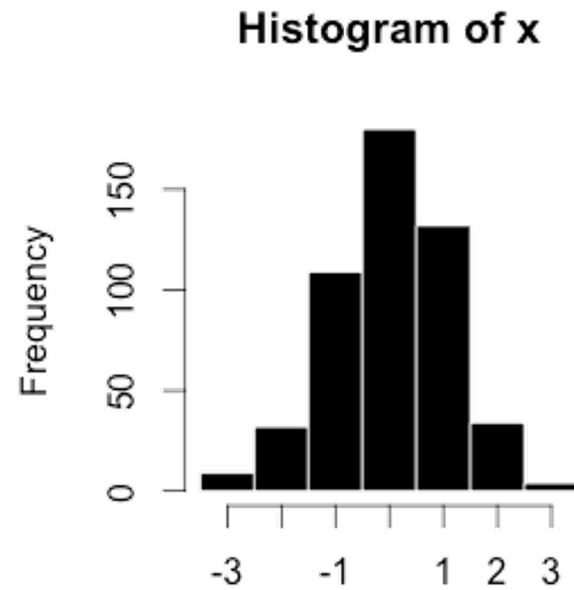
- Non-parametric model
 - Histogram
 - Kernel Density Estimation (KDE)
- Parametric models
 - Gaussian (Normal)
 - Gaussian mixture models (GMM)

Non-parametric Distributions: Histogram

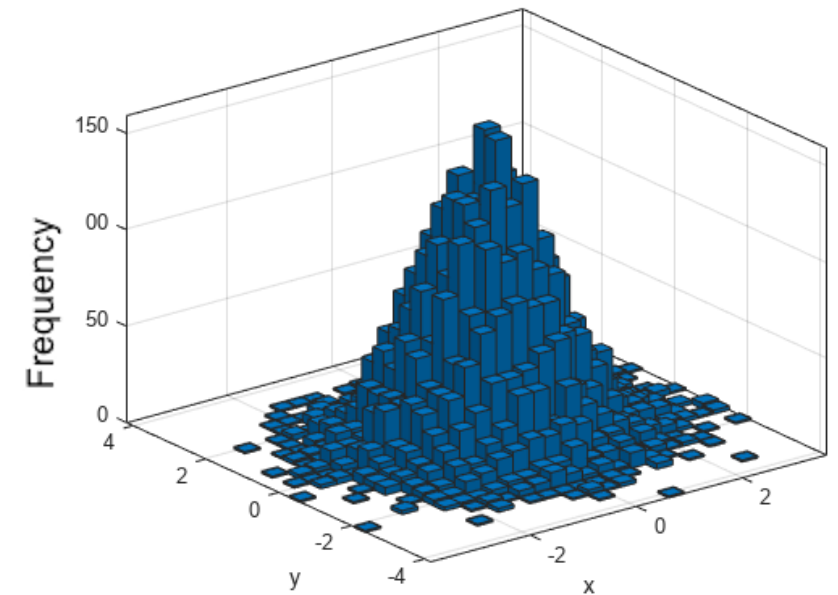
- Histogram: A histogram is an approximate representation of a statistical distribution. The area under a histogram can be normalized and used as a probability distribution function.

$$H(s) = \sum_i \delta(x - x_i)$$

$$\delta(x) = \begin{cases} 1, & \text{if } x = 0 \\ 0, & \text{otherwise} \end{cases}$$



Univariate Histogram



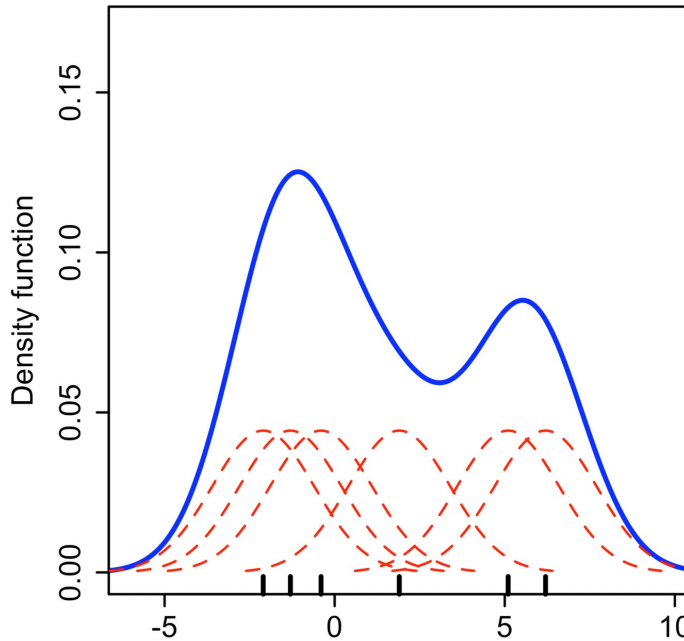
Joint Histogram

Non-parametric Distributions: KDE

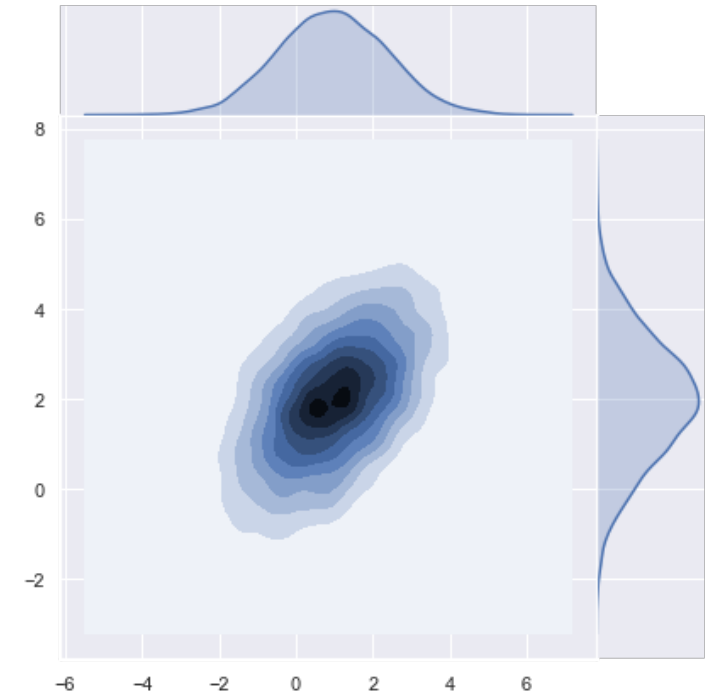
- KDE: Kernel Density Estimation is a popular method of distribution estimation technique from sample data. Formally it is defined as follows:

$$f(x) = \frac{1}{nb} \sum_{i=1}^n K\left(\frac{x - x_i}{b}\right)$$

- $f(x)$ is the KDE function
- n = number of data points
- b = bandwidth
- $K(.)$ = Non-negative symmetric kernel function such as uniform, triangular, Gaussian etc.



Univariate KDE



Joint KDE

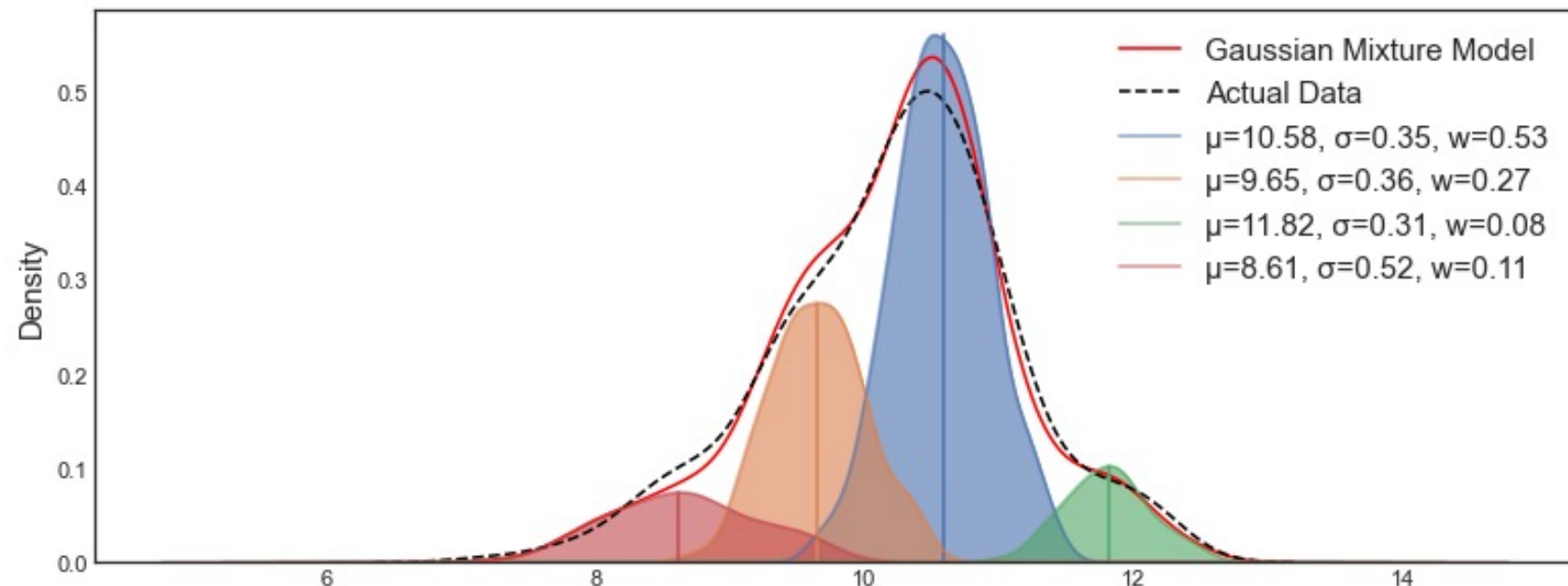
Parametric Distribution: GMM

- Gaussian Mixture Model (GMM): Represent a probability distribution function as a convex combination of multiple Gaussian functions

$$p(X) = \sum_{i=1}^K \omega_i * N(X | \mu_i, \sigma_i)$$

ω = Weights of the Gaussian components

K = Number of Gaussian components in the mixture model

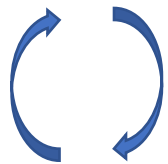


Parameter Estimation Techniques

- Estimation of Gaussian distribution parameters are trivial
 - Maximum Likelihood Estimate (MLE)
 - Same as computing mean and variance
- Estimation of GMM parameters require Expectation Maximization (EM) algorithm
 - Iterative technique to fit GMM parameters
- Incremental schemes for GMM parameter estimation
 - Fast and approximate method to estimate GMM parameters
 - Can model streaming time-varying data

Expectation Maximization (EM) for GMM

- Initialize: means (μ), covariances (Σ), and weights (ω)
- Iterate until convergence:
 - **E-step:** Evaluate posterior probabilities given current parameters



$$\gamma_k^n = \frac{\omega_k \mathcal{N}(x^n | \mu_k, \Sigma_k)}{\sum_{j=1}^K \omega_j \mathcal{N}(x^n | \mu_j, \Sigma_j)}$$

- **M-step:** Update the parameters to maximize the expected log-likelihood of the observed data

$$\omega_k = \frac{N_k}{N} \quad \text{and} \quad N_k = \sum_{n=1}^N \gamma_k^n$$

$$\mu_k = \frac{1}{N_k} \sum_{n=1}^N \gamma_k^n x^n$$

$$\Sigma_k = \frac{1}{N_k} \sum_{n=1}^N \gamma_k^n (x^n - \mu_k) (x^n - \mu_k)^\top$$

- Evaluate log likelihood at the end of each iteration and check for convergence

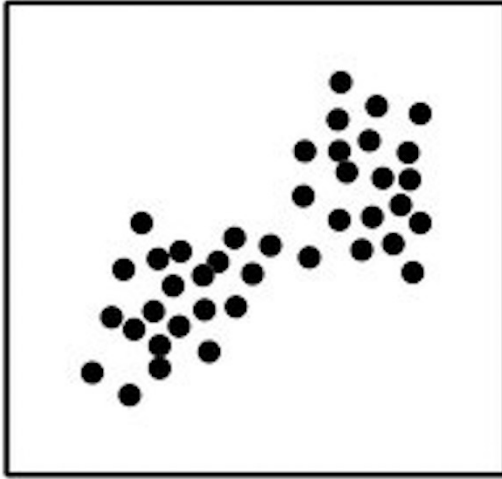
The theory of the method is much more involved!

For a detailed derivation:

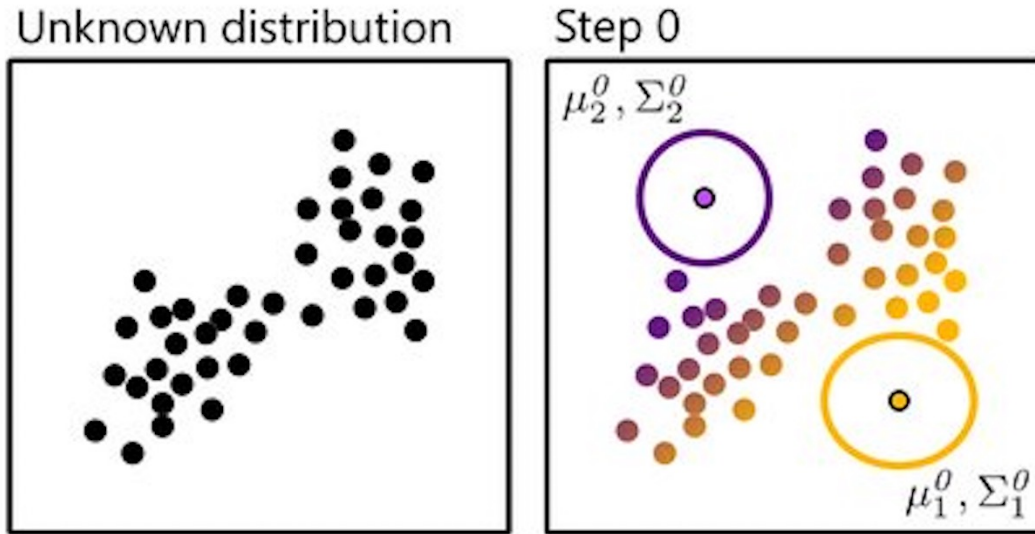
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How EM Algorithm Works

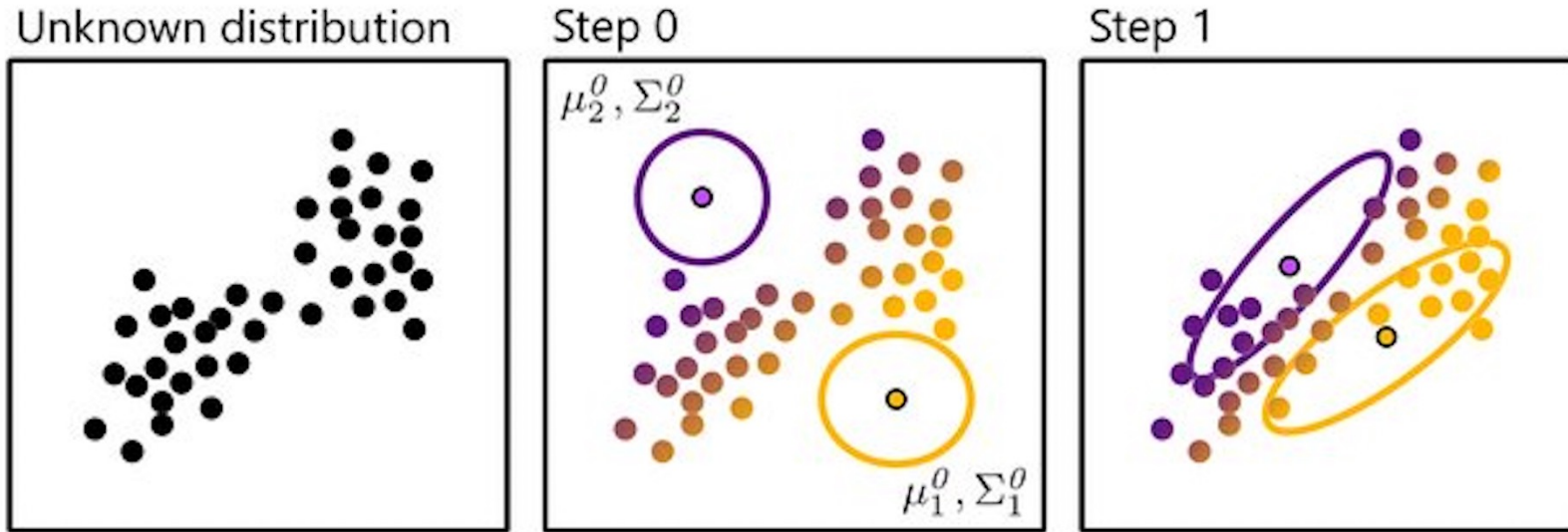
Unknown distribution



How EM Algorithm Works

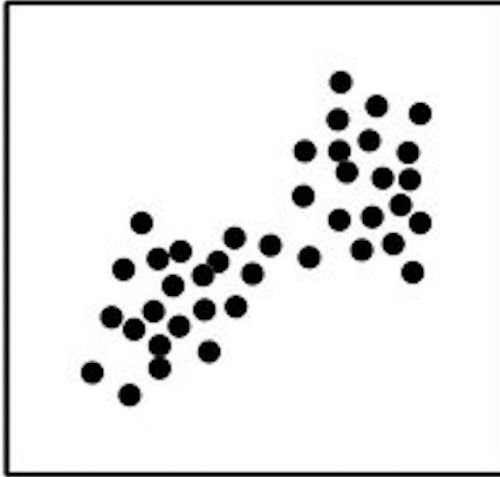


How EM Algorithm Works

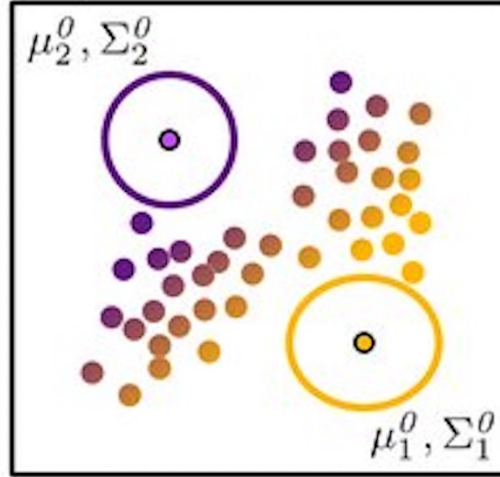


How EM Algorithm Works

Unknown distribution



Step 0



Step 1

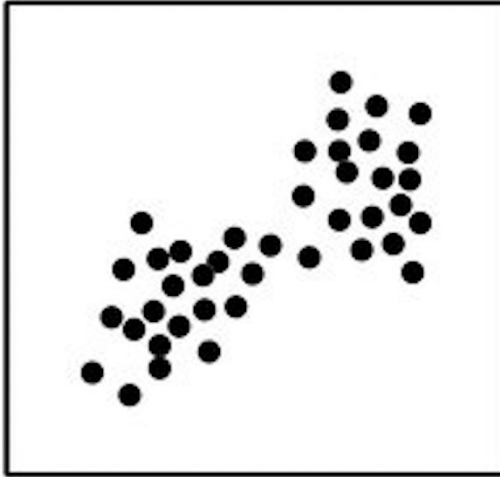


Step 2

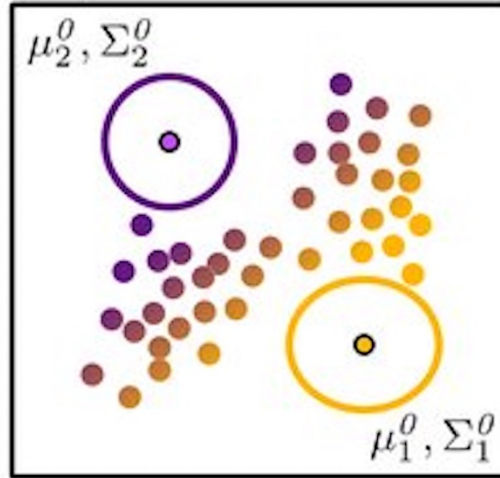


How EM Algorithm Works

Unknown distribution



Step 0



Step 1



Step 2

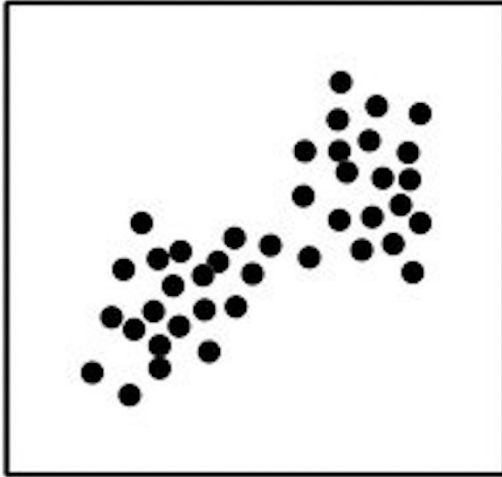


Step 10

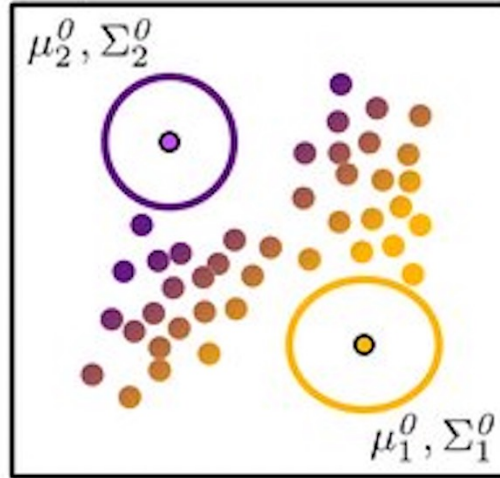


How EM Algorithm Works

Unknown distribution



Step 0



Step 1



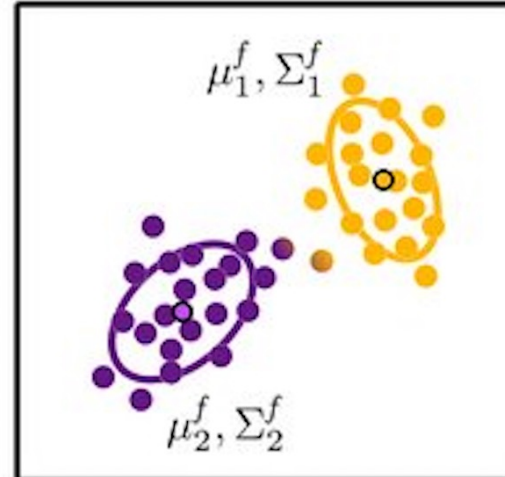
Step 2



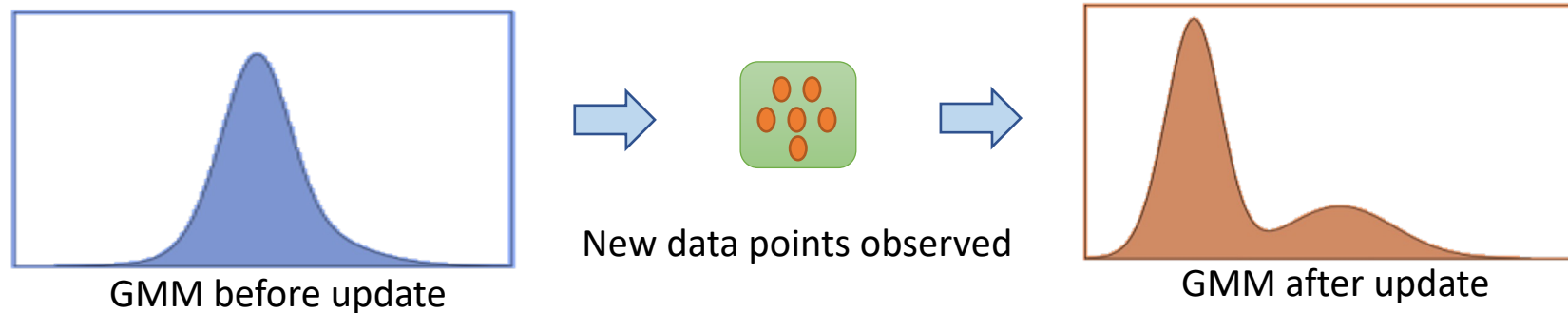
Step 10



Final GMM



Incremental GMM Modeling for Time-varying Data



- Update weights as:

$$\omega_{k,t} = (1 - \alpha)\omega_{k,t} + \alpha * M_{k,t} , \quad M_{k,t} = 1 \text{ for matched dist., } 0 \text{ for others}$$

- Update means and covariances for the matched distribution as:

$$\mu_t = (1 - \rho) \mu_{t-1} + \rho x_t$$

$$\sigma_t^2 = (1 - \rho) \sigma_{t-1}^2 + \rho(x_t - \mu_t)^T(x_t - \mu_t) , \quad \rho = \alpha * N(x_t | \mu_k, \sigma_k)$$

$$\Sigma_{k,t} = \sigma_k^2 I, \text{ where } I = \text{Identity matrix, } \alpha = \text{learning rate}$$